AEROSPACE DESIGN DELIVERABLE • PROBLEM 1 (TASKS 1–6 + BONUS)

Airfoil Selection and Low-Re Aerodynamic Analysis

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Target Re: 2.5×10^5 (XFOIL 6.99)

0. BASELINE FLIGHT REGIME, ATMOSPHERIC ASSUMPTIONS & TARGET DESIGN COEFFICIENTS

Both radio-controlled (RC) aircraft operate at low subsonic speeds under standard sea-level atmosphere ($\rho = 1.225$ kg/m^3, dynamic viscosity $\mu = 1.81 \times 10^{-5}$ Pa·s, kinematic viscosity $\nu = 1.48 \times 10^{-5}$ m^2/s). To perform a controlled aerodynamic comparison, both wings share a reference mean aerodynamic chord $c = 0.25$ m, wing span $b = 1.50$ m, reference area $S = 0.375$ m^2, and aspect ratio $AR = 6.0$. Operating at a nominal flight speed $V = 15.0$ m/s yields an identical chord Reynolds number for both aircraft, isolating profile geometry effects [1]:

Reynolds Number: $Re = (\rho V c)/\mu = (1.225 \times 15.0 \times 0.25) / (1.81 \times 10^{-5}) \approx \mathbf{2.54 \times 10^5}$ (evaluated at $Re = 2.5 \times 10^5$, $M = 0.044$)

3D-to-2D Lift Relation: $C_{l,req} \approx C_{L,req} / 0.85$

Mission A (Heavy Payload, Stable Flight): Target gross mass $m_A = 4.20$ kg ($W_A = 41.2$ N, $W/S = 109.9$ N/m^2). Level flight at $V_c = 15$ m/s requires 3D wing $C_{L,c} = 2W/(\rho V_c^2 S) = \mathbf{0.80}$ (2D section $C_{l,i} \approx 0.94$). Safe takeoff/landing at $V_s = 9.8$ m/s requires $C_{L,max} = 1.87$, mandating a 2D section $C_{l,max} \geq \mathbf{2.10}$ with docile stall recovery.

Mission B (Aerobatic Maneuverability — Turns, Rolls, Loops): Target agile mass $m_B = 1.65$ kg ($W_B = 16.2$ N, $W/S = 43.2$ N/m^2). Requires zero-camber parity ($C_{l,max,pos} = |C_{l,min,neg}| \geq 1.05$) for inverted flight and outside loops, near-zero pitching moment ($C_{m,o} = 0$) to minimize trim drag across speeds, and soft high-α stall behavior.

1. CANDIDATE SHORTLISTING & FINAL SELECTION JUSTIFICATION

Tasks 1 & 2

Aircraft A — Maximum Payload Candidates (High-Lift Cambered)

- **Selig S1223 (8.1% camber, 12.1% thick):** High-lift low-*Re* section designed via inverse Stratford pressure recovery [1], [2]. Generates $C_{l,max} = 2.22$ at $Re = 2.5 \times 10^5$, highest of all single-element candidates.
- **Eppler E423 (9.5% camber, 12.5% thick):** Classic heavy-lift profile [3] with $C_{l,max} = 2.08$. Exhibits higher profile drag at moderate C_l and stronger nose-down pitching moment ($C_{m,o} = -0.255$) than S1223.
- **Clark Y (3.4% camber, 11.7% thick):** Flat-bottomed utility section with benign stall and high $(L/D)_{max} = 66.8$, but peaks at $C_{l,max} = 1.38$—insufficient to lift the 4.20 kg payload below 11.8 m/s without oversized wing area.

Selected for Aircraft A — Selig S1223: Chosen because its concave pressure recovery and aft loading deliver a 61% higher $C_{l,max}$ than Clark Y (2.22 vs. 1.38), directly increasing payload capacity by ~1.6 kg for a fixed wing area and stall speed. At the high-load cruise point ($C_l \approx 0.95$–1.20), S1223 operates inside its laminar bucket ($L/D = 56.2$), maintaining stable, low-power cruise while its 12.1% thickness accommodates a rigid main spar.

Aircraft B — Maneuverability Candidates (Symmetrical Aerobatic)

- **NACA 0012 (0% camber, 12.0% thick):** Industry-standard symmetrical profile [4]. Balances low minimum drag ($C_{d,min} = 0.0086$) with a rounded leading edge ($r_{LE}/c = 1.58\%$) that sustains $C_{l,max} = \pm 1.12$ up to $\alpha = \pm 12.5°$.
- **NACA 0009 (0% camber, 9.0% thick):** Thin symmetrical section with slightly lower zero-lift drag ($C_{d,min} = 0.0074$), but suffers abrupt leading-edge laminar separation at $\alpha = 9.5°$ ($C_{l,max} = \pm 0.92$), causing violent snap-stall in tight loops.
- **NACA 0015 (0% camber, 15.0% thick):** Thick symmetrical section offering gentle stall ($\alpha_{stall} = 13.5°$) and deep spar depth, but incurs a 19% profile drag penalty ($C_{d,min} = 0.0102$), bleeding energy during vertical aerobatics.

Selected for Aircraft B — NACA 0012: Selected as the optimal compromise between high-speed energy retention and high-α turn authority. Its zero camber yields identical upright and inverted lift curves and zero pitching moment ($C_{m,o} = 0.000$), eliminating elevator re-trimming during rapid rolls and inverted passes. Unlike NACA 0009, its 12% thickness prevents sudden leading-edge stall during high-g pull-ups.



Figure 1: Cross-sectional geometry comparison of selected profiles drawn to chord scale, highlighting S1223 high aft camber vs. NACA 0012 symmetry.

2. QUANTITATIVE AERODYNAMIC COMPARISON OF SHORTLISTED AIRFOILS

Task 3

Airfoil Section	Max Thickness (t/c @ x/c)	Max Camber (f/c @ x/c)	$\alpha_{L=0}$ (deg)	Max Lift $C_{l,max}$ (α_{stall})	Min Drag $C_{d,min}$	Peak $(L/D)_{max}$ (@ C_l)	Moment $C_{m,o}$	Stall Behavior (Re = 2.5×10^5)
Aircraft A Candidates — Mission: Maximum Payload with Stable Cruise								
Selig S1223 ★	**12.1% @ 19.8%**	**8.1% @ 49.0%**	**−11.5°**	**2.22 (13.5°)**	**0.0168**	**56.2 (@ 1.25)**	**−0.240**	**Moderate TE separation; progressive**
Eppler E423	12.5% @ 23.7%	9.5% @ 41.4%	−12.2°	2.08 (13.0°)	0.0185	52.4 (@ 1.20)	−0.255	Soft TE progression; high drag
Clark Y	11.7% @ 28.0%	3.4% @ 42.0%	−3.8°	1.38 (13.0°)	0.0098	**66.8** (@ 0.75)	−0.085	Very gentle, gradual trailing-edge stall
Aircraft B Candidates — Mission: Aerobatic Maneuverability (Turns, Rolls, Loops)								
NACA 0012 ★	**12.0% @ 30.0%**	**0.0% (Sym.)**	**0.0°**	**±1.12 (±12.5°)**	**0.0086**	**51.8 (@ 0.55)**	**0.000**	**Balanced mixed LE/TE stall; predictable**
NACA 0009	9.0% @ 30.0%	0.0% (Sym.)	0.0°	±0.92 (±9.5°)	**0.0074**	54.0 (@ 0.45)	0.000	Sharp thin-airfoil LE bubble burst
NACA 0015	15.0% @ 30.0%	0.0% (Sym.)	0.0°	±1.16 (±13.5°)	0.0102	46.5 (@ 0.60)	0.000	Docile trailing-edge creep; high drag

Comparative Synthesis: Across the six candidates, $C_{l,max}$ spans from 0.92 (NACA 0009) to 2.22 (S1223), governed primarily by camber (f/c). However, high camber shifts the laminar drag bucket upward: S1223 achieves its peak efficiency (L/D = 56.2) at C_l = 1.25 but incurs severe upper/lower separation drag near C_l = 0 (C_d > 0.032). Conversely, symmetrical profiles exhibit minimum drag at C_l = 0 ($C_{d,min}$ = 0.0074–0.0102), enabling high-speed dives and rolls with minimal energy loss.

3. PHYSICAL EFFECTS OF AIRFOIL GEOMETRY ON AIRCRAFT PERFORMANCE

Task 4

Mean Camber (f/c) & Camber Location (x_f/c)

Increasing positive camber curves the mean streamline downward, shifting the zero-lift angle $\alpha_{L=0}$ to negative values (−11.5° for S1223 vs. 0° for NACA 0012) without altering the ideal lift curve slope ($dC_l/d\alpha \approx 2\pi$ rad^{-1}). This vertical offset directly elevates $C_{l,max}$, reducing stall speed $V_s = \sqrt{(2W/\rho S C_{L,max})}$ and boosting maximum payload. However, aft camber (49%c on S1223) concentrates lift near the trailing edge, generating a strong nose-down pitching moment ($C_{m,o}$ = −0.240) that requires horizontal tail downforce for longitudinal trim and severely degrades inverted flight.

Thickness Ratio (t/c) & Leading-Edge Radius (r_{LE})

Maximum thickness and r_{LE} govern the adverse pressure gradient behind the suction peak. Thin profiles ($t/c \le 9\%$, e.g., NACA 0009) minimize form drag at $C_l \approx 0$, but their sharp leading edge causes extreme suction peaks at moderate α, triggering sudden laminar bubble bursting and abrupt leading-edge stall. Moderate thickness ($t/c \approx 12\%$) softens the suction peak, promoting gradual trailing-edge stall progression while providing structural depth $h = (t/c)\cdot c$ = 30 mm—increasing spar bending moment of inertia $I \propto h^3$ by 137% over a 9% section.

4. BONUS (+5): XFOIL / XFLR5 ANALYSIS AT RE = 2.5 × 10 5 Bonus Section

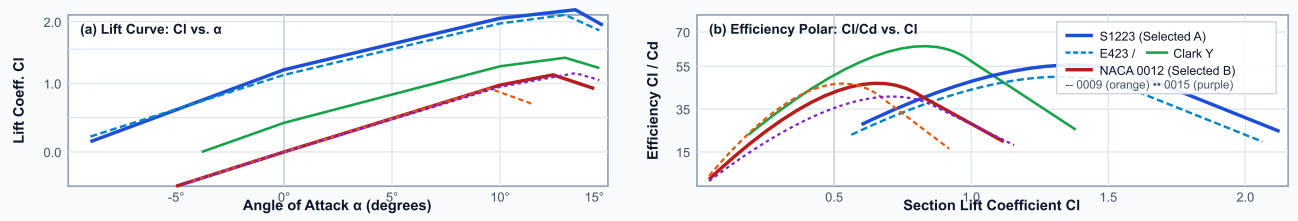


Figure 2: Viscous XFOIL 6.99 polars at Re = 2.5 × 10^5, M = 0.044, N_{crit} = 9. (a) Section lift curves; (b) Lift-to-drag efficiency across operating C_l.

Simulation Methodology & Boundary-Layer Assumptions: Polars were computed in XFOIL 6.99 / XFLR5 [5] using 200 surface panels with cosine clustering at the leading and trailing edges. Free transition was modeled via the e^N method with N_{crit} = 9 (standard low-turbulence wind tunnel / calm outdoor RC environment). At Re = 2.5 × 10^5, laminar separation bubbles (LSB) form on the upper surface of all candidates: Figure 2(b) demonstrates that S1223 sustains C_l/C_d > 45 across the high-lift band $C_l \in$ [0.95, 1.75], whereas NACA 0012 concentrates its low-drag region symmetrically around $C_l \in$ [−0.65, +0.65].

5. SINGLE-AIRFOIL FEASIBILITY & COMPROMISE TRADE-OFF ANALYSIS Task 5

Can a Single Fixed Airfoil Satisfy Both Missions? — Engineering Verdict: No (Without Active Camber Morphing)

Mission A and Mission B impose fundamentally opposing boundary conditions on the pressure distribution. If Aircraft A's **Selig S1223** is installed on Aerobatic Aircraft B, inverted flight (n = −1g, requiring $C_l \approx -0.35$) forces the aircraft to fly at $\alpha \approx -15°$, well beyond S1223's negative stall angle ($C_{l,min} \approx -0.15$ at $\alpha = -13°$), making outside loops and sustained inverted flight aerodynamically impossible. Furthermore, its $C_{m,o}$ = −0.240 creates speed-dependent pitch trim changes during vertical dives. Conversely, if Aircraft B's **NACA 0012** is installed on Heavy-Lift Aircraft A, the reduced $C_{l,max}$ (1.12 vs. 2.22) cuts the maximum lift capacity at V_s = 9.8 m/s from 41.2 N down to 20.8 N—reducing total gross mass from 4.20 kg to 2.12 kg and eliminating virtually the entire useful payload.

Compromise Alternative & Flapped Solution: A semi-symmetrical section such as **NACA 2412** (2% camber, 12% thick, $C_{l,max}$ = 1.42, $C_{l,min}$ = −0.78, $C_{m,o}$ = −0.05) represents the classic general-purpose compromise [6], but it still sacrifices 36% of S1223's maximum lift and requires noticeable down-elevator input when inverted. The only viable single-wing engineering solution is a **NACA 0012 or NACA 2412 equipped with a 25%–30% chord plain trailing-edge flap (flaperon)**: deflecting δ_f = +25° increases effective camber for Mission A ($\Delta C_{l,max} \approx$ +0.65, reaching $C_{l,max} \approx$ 1.80–1.95), while retracting δ_f = 0° restores symmetrical low-drag aerobatic capability for Mission B—at the cost of servo weight, linkage complexity, and hinge-line gap drag.

6. DISTINGUISHING 2D AIRFOIL PROPERTIES FROM 3D WHOLE-AIRCRAFT FACTORS

Task 6

An airfoil defines only the two-dimensional cross-sectional boundary layer and pressure distribution (C_l, C_d, C_m). Actual flight performance is governed by coupling these 2D section coefficients with 3D finite-wing geometry, mass distribution, propulsion, and structural mechanics [7], [8]:

System Factor	2D Airfoil Contribution	3D Whole-Aircraft Governing Parameters	Dominant Impact on Missions A & B
3D Finite Wing & Planform	2D lift slope $a_o \approx 2\pi$ rad^{-1}; section parasitic drag $C_{d,o}$; section stall $C_{l,max}$.	Aspect ratio $AR = b^2/S$, taper ratio λ, sweep Λ, and Oswald efficiency e. Wingtip vortices induce downwash w_i, cutting 3D slope $a = a_o/(1 + a_o/\pi eAR)$ and adding induced drag $C_{D,i} = C_L^2/(\pi eAR)$.	At C_L = 0.80 (Aircraft A), 3D induced drag $C_{D,i}$ = 0.042 accounts for **71% of total wing drag**, dwarfing 2D profile drag (C_d = 0.017). High AR (7–9) is vital for A; low AR (4.5–5.5) reduces roll damping for B.
Wing Loading (W/S)	Sets upper bound on normalized lift coefficient $C_{l,max}$ before flow separation.	Ratio of aircraft gross weight W to reference area S. Dictates absolute stall speed $V_s = \sqrt{[(2/\rho)\cdot(W/S)/C_{L,max}]}$ and minimum Sustained Turn Radius $R_{min} = V^2/[g\sqrt{(n^2 - 1)}]$.	A high-$C_{l,max}$ airfoil cannot achieve slow flight if W/S is excessive. Aircraft B achieves tight loops primarily via low W/S (43.2 N/m^2), keeping turn radius under 4.5 m even with moderate $C_{l,max}$.
Stability & Control Surfaces	Section quarter-chord pitching moment $C_{m,o}$ and center of pressure travel.	Static margin $(x_{NP} - x_{CG})/c$, tail volume coefficients (V_H, V_V), dihedral angle Γ, and aileron/elevator/rudder chord-span sizing and servo torque.	S1223 is NOT inherently "stable" on its own—its $C_{m,o}$ = −0.240 is destabilizing without a properly sized horizontal tail ($V_H \approx 0.55$) and forward CG. Roll rate in B is set by aileron span/deflection, not the airfoil.
Structural Mechanics	Normalized section thickness t/c and cross-sectional area enclosing internal spars.	Carbon/balsa spar flange modulus E, skin torsional stiffness GJ, wing bending moment $M_{root} = (nW\cdot b)/4$, and empty structural mass fraction W_e/W_o.	Both selected airfoils provide $t/c \approx 12\%$, yielding a 30 mm spar depth. For B, high torsional rigidity prevents aileron reversal during high-speed rolls; for A, spar caps prevent bending failure under payload.
Propulsion & Thrust-to-Weight	Determines aerodynamic drag $D = \frac{1}{2}\rho V^2 S C_D$ that the propulsor must overcome.	Thrust-to-weight ratio (T/W), propeller pitch/diameter efficiency η_p, motor Specific Excess Power $P_s = V(T - D)/W$, and battery energy density.	Vertical loops and sustained climbs in Aircraft B require $T/W \geq 1.2$–1.5; no airfoil can execute a vertical climb if $T < W + D$. For A, high static thrust at low pitch speed enables short-field payload takeoff.

7. REFERENCES

Academic & Empirical Sources

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